

PROJECTREPORT

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfillment of the requirement of degree of

B.Tech CSE(2023 – 2027)



By

Roll Number : CS-23411536

Name of Student: shubham

kushwaha Batch: 2024-2027

CANDIDATE'S DECLARATION

I, shubham kushwaha do hereby declare that the internship report titled “**Topic**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name : shubham
kushwaha Roll No. : CS-
23411536

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Colourbar Communications for providing me the opportunity to undertake my internship as a Data Science Intern within their Artificial Intelligence (AI) department. This internship gave me valuable exposure to real-world data science practices and allowed me to apply my academic learning to a live organisational project.

I am especially thankful to Mr. Nitin Narula, Partner, Colourbar Communications, for facilitating this internship and for the guidance extended to me during its course.

I am also grateful to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout this internship and in the preparation of this report.

Finally, I would like to thank my parents and family for their constant encouragement and blessings, without which this endeavour would not have been possible.

Student Name: shubham kushwaha

Roll No.: CS-23411536

Internship Completion Certificate

The internship completion certificate issued by Colourbar Communications, certifying the successful completion of the internship, is reproduced below.



Dated 25th July 2026

TO WHOM SO EVER IT MAY CONCERN

This is to certify that Mr. Ashwin Saxena has successfully completed his six-week remote internship as a Data Science Intern with Colourbar Communications, from June 1, 2025, to July 13, 2026. During his tenure, he was an integral part of the Artificial Intelligence (AI) department.

During the internship, Mr. Ashwin Saxena actively contributed to the "Potential Client Analysis and Automation" project. His primary focus was leveraging data-driven strategies to identify high-value clients and engineering automated workflows to enhance organizational decision-making and operational efficiency.

Key Contributions and Achievements:

- **Data Management:** Executed end-to-end data extraction, cleaning, and pre-processing from diverse internal repositories to build comprehensive client profiles.
- **Exploratory Analysis:** Conducted detailed Exploratory Data Analysis (EDA) to uncover critical market trends, behavioural patterns, and key drivers of client acquisition.
- **Predictive Analytics:** Developed and deployed machine learning algorithms to score, rank, and segment potential client leads based on dynamic business parameters.
- **Workflow Automation:** Authored robust Python scripts and automated data pipelines, significantly optimizing the client scoring workflow and minimizing manual processing time.
- **Data Visualization:** Designed intuitive, interactive dashboards to communicate complex analytical insights to cross-functional stakeholders effectively.

Throughout his time with us, Mr. Ashwin Saxena demonstrated exceptional analytical aptitude, advanced problem-solving skills, and a practical mastery of core data science principles. His dedication and technical contributions were highly valuable to the team's success.

We extend our sincere appreciation for his efforts and wish him the very best in his future academic and professional pursuits.

For Colourbar Communications


Nitin Narula
Partner

44, Community Centre, Upper Ground Floor, Naraina Industrial Area, Phase -I, New Delhi - 110028, (Above OBC Bank)
☎ Tel: 011-42455108 ✉ E-mail: colourbarcommunications@gmail.com 🌐 Website: www.colourbar.in

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4. Project Description

4.1 Introduction

As part of the curriculum for the degree of Bachelor of Technology in Computer Science

and Engineering (Graphics and Gaming) at IILM University, Greater Noida, U.P., I, shubham kushwaha (Roll No. CS-23411536, Section 4CSE11, 6th Semester), undertook an industrial internship to gain practical, industry-oriented experience in the field of data science.

The internship was completed with Colourbar Communications as a Data Science Intern, working within the organisation's Artificial Intelligence (AI) department. The internship was conducted in remote mode over a duration of six weeks, from June 1, 2025, to July 13, 2025.

During this period, I was involved in the project titled “Potential Client Analysis and Automation,” which focused on using data-driven strategies to identify high-value clients and on engineering automated workflows to enhance organisational decision-making and operational efficiency.

This report documents the organisation with which the internship was carried out, the problem that the project addressed, the objectives and scope of the work, the technologies used, the methodology followed, and the outcomes expected from the project. The report has been prepared in accordance with the format prescribed by the School of Computer Science and Engineering, IILM University.

4.2 Organisation Profile

The internship was undertaken at Colourbar Communications, whose registered office is

located at 44, Community Centre, Upper Ground Floor, Naraina Industrial Area, Phase-I, New Delhi – 110028.

Contact details of the organisation, as stated on its official communication, are as follows:

- Telephone: 011-42455108

- Email: colourbarcommunications@gmail.com

Within the organisation, I was placed in the Artificial Intelligence (AI) department, which is responsible for data-driven initiatives supporting the organisation's business operations.

4.3 Problem Statement

Organisations that rely on manual evaluation of prospective clients often face challenges in efficiently identifying which leads are most likely to convert into high-value business relationships. Manual client evaluation tends to be time-consuming and does not scale well as the volume of prospective client data grows.

The core problem addressed during this internship was the need for a data-driven and automated approach to identify high-value potential clients and to streamline the associated decision-making process, thereby improving the organisation's operational efficiency.

4.4 Project Objectives

Based on the work carried out during the internship, the key objectives of the project were:

- To extract, clean, and pre-process client-related data from diverse internal repositories in order to build comprehensive client profiles.
- To conduct exploratory data analysis (EDA) to uncover market trends, behavioural patterns, and key drivers of client acquisition.
- To develop and deploy machine learning-based methods to score, rank, and segment potential client leads according to business-defined parameters.
- To automate the client scoring workflow through scripting and data pipelines, reducing manual processing effort.
- To design interactive dashboards that communicate analytical insights clearly to cross-functional stakeholders.

4.5 Scope of the Project

The scope of the project was limited to the Artificial Intelligence (AI) department of

Colourbar Communications and was carried out over a six-week remote internship period. The work centred on the “Potential Client Analysis and Automation” initiative and covered the stages of data collection, data preparation, analysis, predictive modelling, workflow automation, and visualisation of results for internal stakeholders.

The project scope did not extend to organisation-wide deployment or long-term maintenance of the solution beyond the internship period, as such details are outside the scope of the information provided in the internship documentation.

4.6 Technologies and Tools Used

The following table summarises the technologies and tools associated with the project,

based on the internship documentation:

Table 4.2: Technologies and Tools Used

Category	Tool / Technology
Programming Language	Python (used for scripting and workflow automation)
Data Analysis	Exploratory Data Analysis (EDA) techniques
Predictive Modelling	Machine learning algorithms for lead scoring, ranking, and segmentation
Automation	Automated data pipelines and Python scripts
Visualisation	Interactive dashboards for stakeholder reporting

4.7 System Architecture

Figure 4.1 below presents the proposed system architecture reflecting the general flow of work carried out during the internship — from raw client data to actionable business insight.

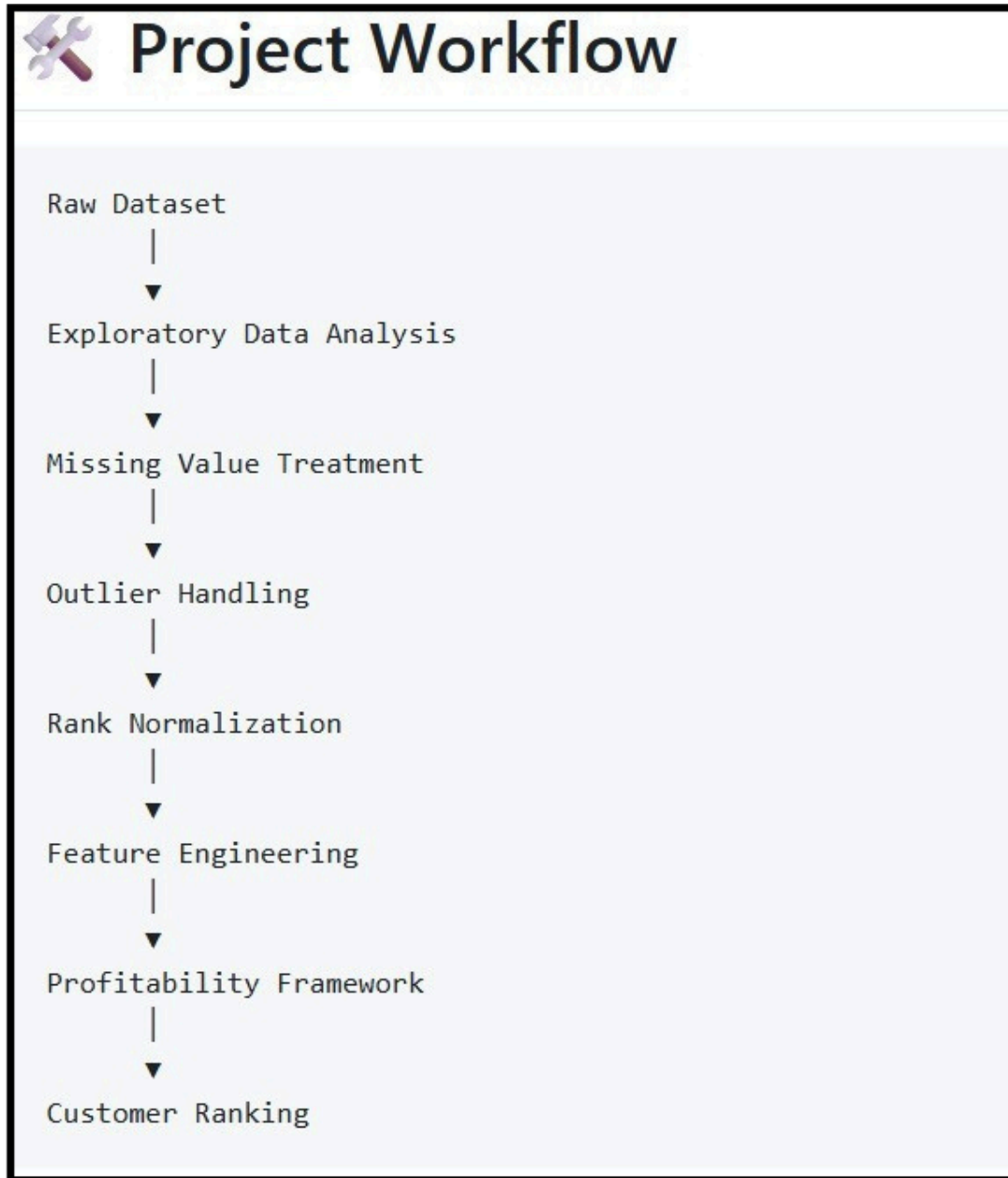
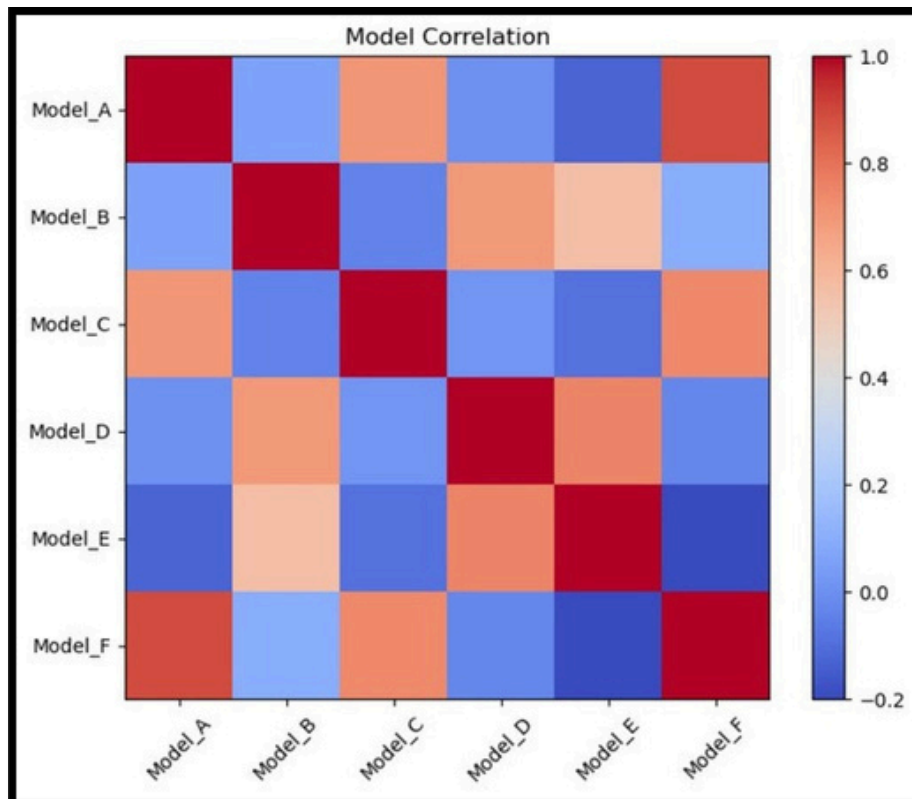


Figure 4.1: Proposed System Architecture for Client Analysis and Automation



```
# =====
# SECTION 0 - SETUP & CRITICAL FIX
# =====

import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd

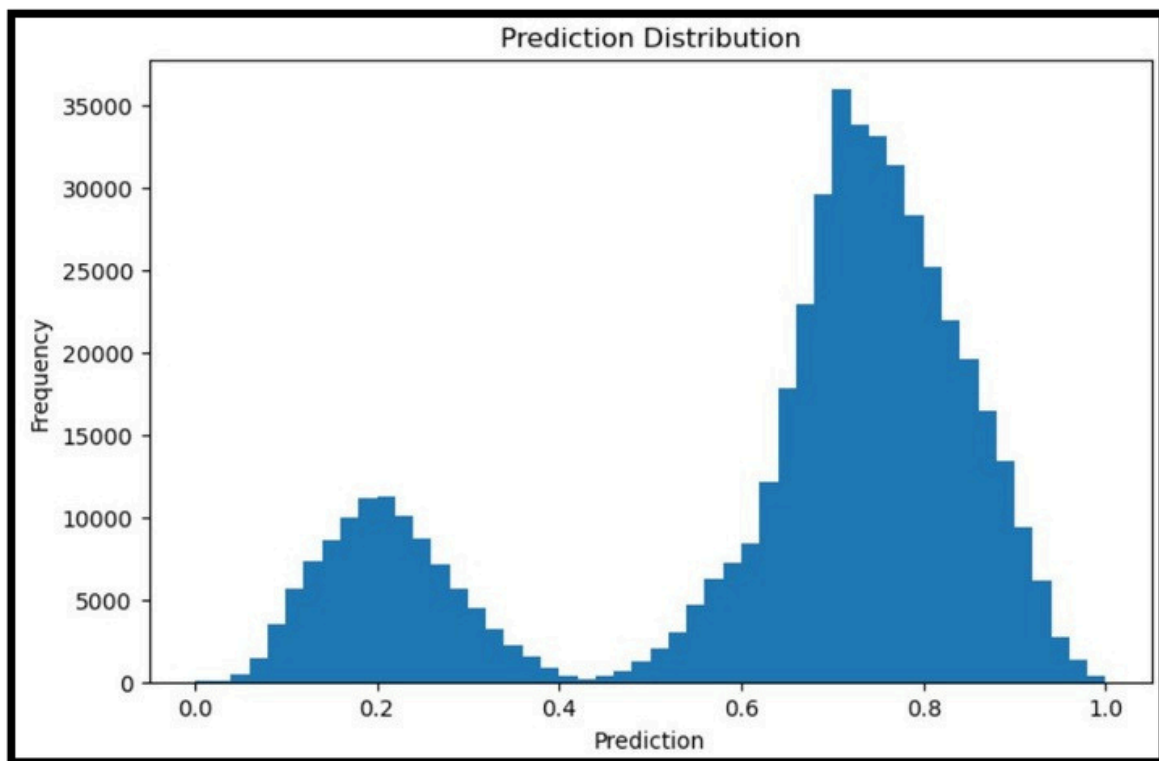
from scipy.stats import skew, kurtosis, rankdata

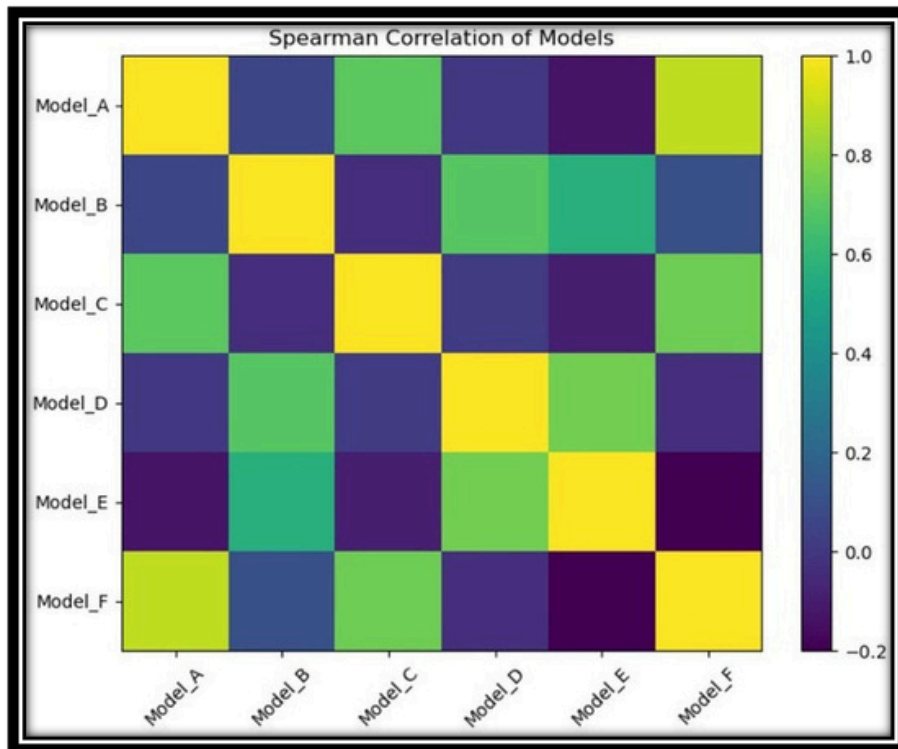
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import QuantileTransformer
from sklearn.decomposition import PCA
from sklearn.cluster import AgglomerativeClustering
from sklearn.mixture import GaussianMixture
from sklearn.ensemble import IsolationForest

import scipy.cluster.hierarchy as sch
import matplotlib.pyplot as plt

# Display settings
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", 100)
pd.set_option("display.float_format", lambda x: f"{x:,.6f}")
```

As documented in the internship certificate, the workflow broadly proceeds from internal data repositories, through a data cleaning and pre-processing stage, into exploratory analysis, machine learning-based scoring, an automated pipeline layer, and finally into a dashboard layer for stakeholder consumption.





4.8 Methodology

The methodology followed during the internship, as reflected in the internship

completion certificate, consisted of the following stages:

- Data Management: End-to-end data extraction, cleaning, and pre-processing from diverse internal repositories to build comprehensive client profiles.
- Exploratory Analysis: Detailed exploratory data analysis (EDA) to uncover critical market trends, behavioural patterns, and key drivers of client acquisition.
- Predictive Analytics: Development and deployment of machine learning algorithms to score, rank, and segment potential client leads based on dynamic business parameters.
- Workflow Automation: Authoring of Python scripts and automated data pipelines to optimise the client scoring workflow and minimise manual processing time.
- Data Visualisation: Designing interactive dashboards to communicate analytical insights to cross-functional stakeholders.

Table 4.1: Summary of Key Contributions during Internship

Contribution Area	Description
Data Management	Extraction, cleaning, and pre-processing of client data
Exploratory Analysis	Identification of market trends and client acquisition drivers
Predictive Analytics	ML-based scoring, ranking, and segmentation of leads
Workflow Automation	Python-based scripts and automated data pipelines
Data Visualisation	Interactive dashboards for stakeholder reporting

4.9 Expected Outcomes

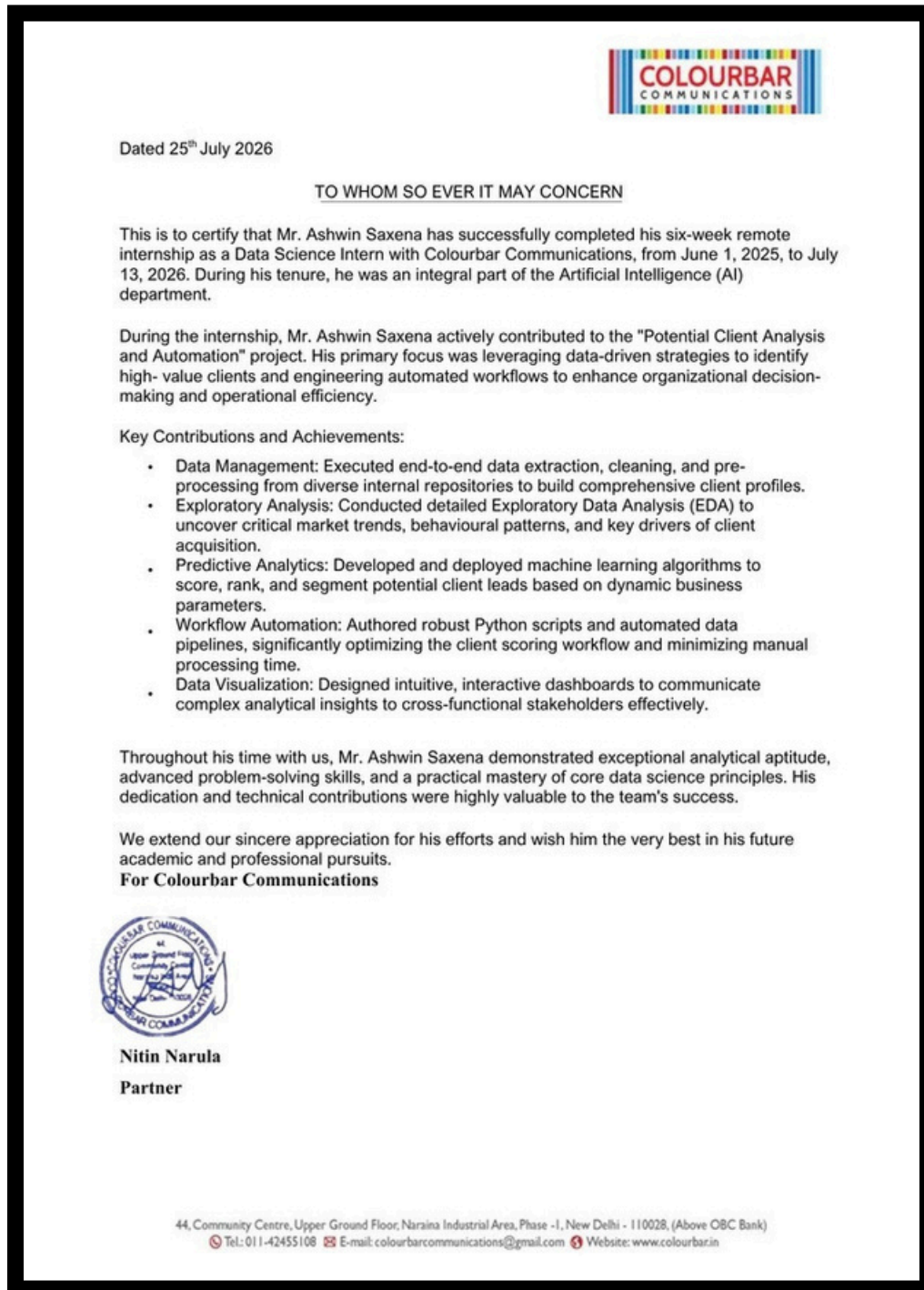
Based on the internship documentation, the project was expected to result in the following outcomes:

- A more efficient, data-driven process for identifying high-value potential clients.
- An optimised client scoring workflow with reduced manual processing time, achieved through workflow automation.
- Improved organisational decision-making, supported by interactive dashboards that present analytical insights to stakeholders.
- Enhanced understanding of client acquisition patterns through exploratory data analysis.

The internship certificate notes that these contributions were considered valuable to the team, and that the intern demonstrated strong analytical aptitude and problem-solving skills. No specific quantitative metrics (e.g., percentage improvement in processing time or lead conversion) were stated in the certificate, and none have been assumed or fabricated for this report.

4.10 Certificates of Completion and Communication Proof

A copy of the internship completion certificate issued by Colourbar Communications has been included in Chapter 3 of this report.



5. Bibliography / References

- 1.ColourbarCommunications, “Internship Completion Certificate – shubham kushwaha, Data Science Intern,” dated July 25, 2026.
2. School of Computer Science and Engineering, IILM University, Greater Noida, U.P., Internship Report Format Guidelines.